Lightweighting with Additive Manufacturing 271

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Class Outline

Lightweighting in Manufacturing
Lightweighting with Additive Manufacturing
Additive Manufacturing Methods for Lightweighting
Industrial Applications of Lightweighting
DFAM and Lightweighting
Review: Lightweighting with AM
Topology Optimization
Lattice Structures
Advantages & Disadvantages of Lightweighting
Review: Lightweighting with AM

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Lightweighting in Manufacturing

Lightweighting is an increasingly popular practice in manufacturing. Lightweighting reduces the weight of parts while ensuring that the parts' functions remain the same. This serves two primary purposes. First, reducing the weight of a product means it requires less energy to move it, both during shipping and during the part's use. Second, using fewer or lighter-weight materials reduces costs and uses fewer resources.

Lightweighting involves continuous improvement as manufacturers regularly seek to further reduce the weight of their products. What was once considered light soon becomes an area of potential improvement. Lightweighting begins as a design strategy. Engineers constantly test new designs for achieving lighter weights without loss of functionality. Lightweighting a part can also affect the design of other parts or components. For example, shifting from steel to a lighter aluminum automobile body may require other changes to the car's design, such as updating paints, adhesives, and welding processes.

Replacing steel parts with aluminum is an example of the popular lightweighting method, material substitution. Lightweighting parts by substituting plastic or carbon fiber instead of metals, for example, has been a long-standing trend, particularly in automotive manufacturing. Recently, however, automakers and other manufacturers are recognizing that material substitution alone may not be enough to reach their lightweighting goals. Additive manufacturing (AM) offers more flexibility than material substitution or traditional manufacturing methods, so manufacturers are increasingly using AM methods to lightweight parts.

This material is based upon research supported by, or in part by, the U.S. Office of Naval Research under award number N00014-18-1-2881 led by the National Center for Defense Manufacturing and Machining (NCDMM).

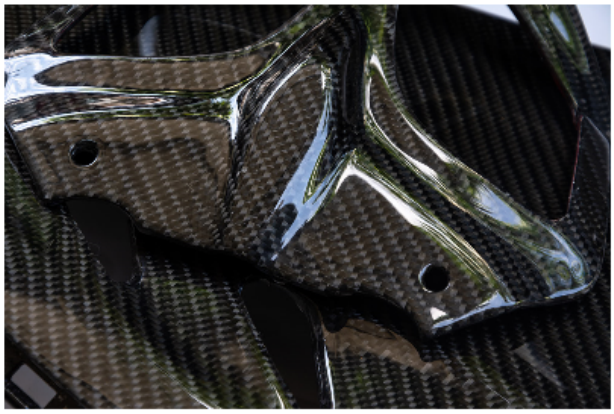

Figure 1: An auto part made of lighter-weight carbon fiber materials.

Figure 2: An additive manufacturing process.

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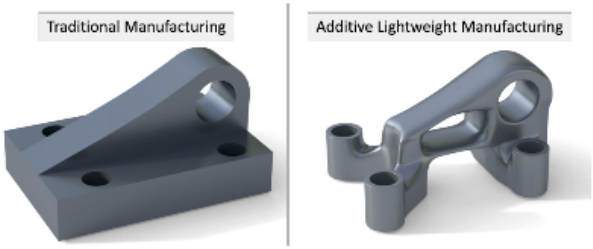
Lightweighting with Additive Manufacturing



Today, several industries use AM to develop and produce lighter parts. The design and build processes of AM allow for lightweighting to occur at multiple steps in the manufacturing process. Software for AM assists in identifying material reduction opportunities in the design phase. AM build processes can produce structures with less density than traditional processes.

In additive manufacturing, an AM machine builds a finished part by stacking and joining layers of material. AM parts often have improved complexity and functionality compared to parts manufactured by traditional processes like machining. Unlike traditional manufacturing, which is limited by things like the size and shape of cutting tools, AM can lay down layers in unique, precise ways. Rather than starting with a solid block of raw material that must be cut or shaped, AM processes can deposit material only where it is necessary, which gives designers much more flexibility to reduce part material.

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Additive Manufacturing Methods for Lightweighting

Directed energy deposition uses a laser or other source of focused thermal energy to melt and fuse materials in order to create a part layer. During DED, a deposition head focuses and directs thermal energy at a base, which can be a build platform or a substrate. In the deposition head, there is also a nozzle that feeds build material into the thermal energy. When the energy source and build material contact one another at the base, a melt pool forms. As the deposition head moves across the base in the shape of the desired part, the melt pool travels with it, creating a material layer that gradually solidifies upon removal of the energy source. DED creates parts by repeatedly passing over the base to build up layers of material.

Material jetting uses ultraviolet light (UV light) to cure photopolymer build material. During material jetting, one or more print heads move horizontally over the build platform to deposit a layer of build material in the shape of a part. The build platform then lowers, and the print heads deposit a new material layer onto the previous layer. Each layer solidifies and binds together. This process repeats until the part is complete.

Powder bed fusion uses focused thermal energy to selectively fuse layers of powdered build material. During PBF, a roller dispenses a layer of powder onto a build platform within a powder bed. An image projection module focuses thermal energy and projects it, moving it in the shape of a part across the powder layer. Contact with the thermal energy fuses the powder particles together and forms a part layer. Then, the roller dispenses a fresh layer of powder over the first, and the process repeats until the part is complete.

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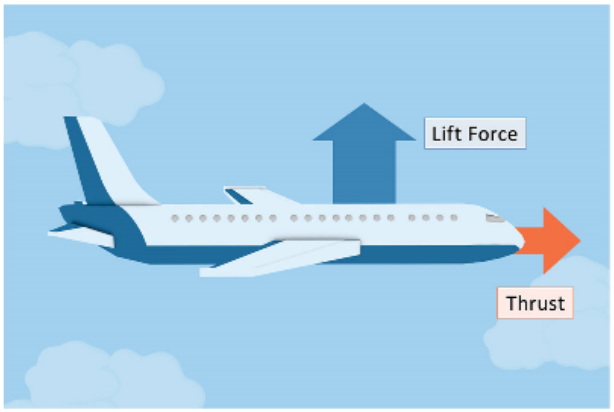
Industrial Applications of Lightweighting

Lightweighting is a top priority in the automotive industry. The goal of reducing overall vehicle weight is to improve fuel efficiency and meet the needs of newer electric vehicles. Under most circumstances, a ten percent reduction in vehicle weight can lead to an eight percent improvement in fuel efficiency. However, lightweighted parts must still meet the same rigorous safety and performance standards as traditional parts.



The aerospace industry also considers fuel efficiency a prime concern. Reducing the mass of aircraft can reduce fuel consumption because a lower mass requires less lift force and thrust during flight. In addition to reducing an aircraft's carbon footprint and operation costs, lightweighting can help to achieve flight performance improvements such as increased safety, better acceleration, and higher structural strength.

In the medical industry, equipment must provide faster, more accurate, and easier-to-manage ways for care professionals to treat patients.

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Lightweighting improves the durability and functionality of components for patient monitoring equipment, advanced robotics, prosthetics, replacement joints, dental implants, and surgical tools. Lighter products can also have increased portability and accuracy.

Energy industries also see increased efficiency with lightweighting. For example, to generate wind power using wind turbines, manufacturers use extremely lightweight materials to create long rotating blades. When the wind rotates the blades, they transform the wind energy into rotating energy that is then transformed into electricity. High wind speeds and the size of the blades place the blade material under high stress, which can lead to performance issues or failure. Lighter weight blades have less inertia, and thus experience less stress when turned by the wind.

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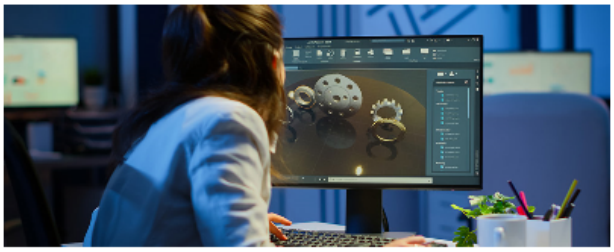
DFAM and Lightweighting



Design for additive manufacturing (DFAM) refers to designing a part or a product for effective and efficient AM. DFAM includes system design, which considers which parts should be manufactured using AM and what the parts should look like.

Most essential to lightweighting processes is part design, a stage of the DFAM process that determines the best geometric design for a single part based on performance requirements, such as whether or not the part must support weight or resist impacts. Designers use CAD software to create and virtually test models of a part, including removing material for lightweighting purposes. When a designer removes material at the CAD level, it is critical that they confirm that the part possesses all the necessary properties, such as strength and safety.

Reducing the mass of a part impacts every other mechanical aspect of its design. Without careful consideration, lighter parts may not meet the same tolerances as their heavier counterparts. Manufacturers must ensure that lightweighted products still meet safety, durability, and performance standards.

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Review: Lightweighting with AM

1. Material reduction is a process that swaps lighter materials for heavier ones. **FALSE**
2. Lightweighting with AM begins at the design level. **TRUE**
3. Reducing the mass of a structure can impact its strength and durability. **TRUE**
4. DFAM seeks the ideal material layout without sacrificing strength. **TRUE**
5. Energy industries require heavy materials and are looking to add weight to parts and products. **FALSE**
6. Additive manufacturing builds parts that are less complex than those made by traditional manufacturing. **FALSE**

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Lesson: 7/10

Topology Optimization
Because hollowing or eliminating areas of a part can impact its strength and durability, designers must test product designs using simulation software. Traditional simulation tools provide details on digital part models, including areas of stress or strain. Lightweighting often requires a specific type of simulation software that can calculate topology optimization (TO). Topology optimization software uses a mathematical method to optimize a part's material layout. Designers enter aspects of the part's intended use, such as the loads it will bear, and the software calculates exactly how much material to use in different areas of the part to meet those requirements. TO can recognize the sections of a part that are the least necessary to its overall strength and recommend eliminating those sections. This speeds up the design process significantly and greatly reduces the time necessary to test physical parts and refine part design prior to production.

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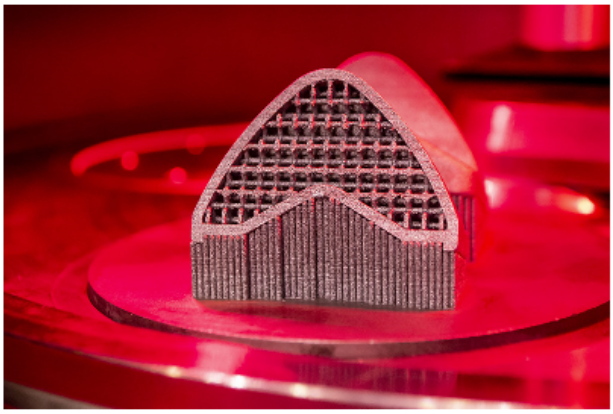
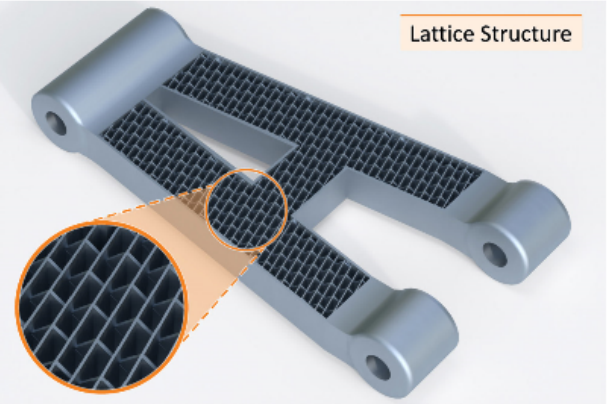
Lattice Structures





A lattice is made of repeated unit cells. These cells can take a variety of shapes and sizes, and they may be repeated in many different ways. A uniform lattice has identical cells repeated in every direction, while a variable lattice has cells of varying size or spacing in different directions. Both kinds of lattice structures can maintain internal support in an AM part, which means that the part can be lighter weight while retaining strength. The weight savings improve the part's strength-to-weight ratio.

A benefit of lattice structures is they can be additively manufactured faster, using less material and requiring less build time than fully dense, solid components. This improved material use may allow cost-effective use of higher-quality, more expensive materials. Since traditional machining operations start with a solid block and cut material away to create a part, creating a lattice structure would not reduce the amount of material required. However, AM can build lattices without significant waste or excess up-front material expense.

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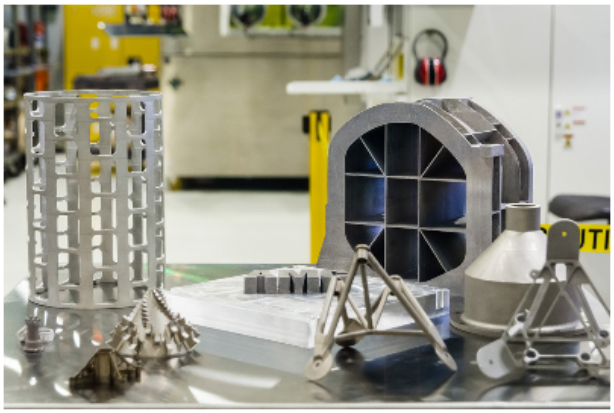
Advantages & Disadvantages of Lightweighting

Additively manufactured parts made up of multiple materials.

AM processes remove some of the extra material required in traditional manufacturing. During production, the AM machine deposits or fuses material only where it is functionally necessary, and unused materials leftover from builds can often be reclaimed and reused in future builds. Lightweighting with AM thus requires less build material overall, which reduces material costs. Additionally, AM can build solid parts from multiple materials. This means that AM can deposit expensive materials only where needed instead of building the whole part from them, which also saves material costs.

Lightweighting makes products that require less energy to ship.

Lightweighting results in parts and products that use less energy, making them more environmentally friendly, which is a concern among many conscious consumers and company leaders facing environmental regulations. Transporting and shipping parts and products of lighter weight also saves energy resources.



Complex geometric structures in AM parts maintain strength.

Lightweighting with AM results in parts that are often at least as strong as heavier, traditionally manufactured parts. This is a benefit of the

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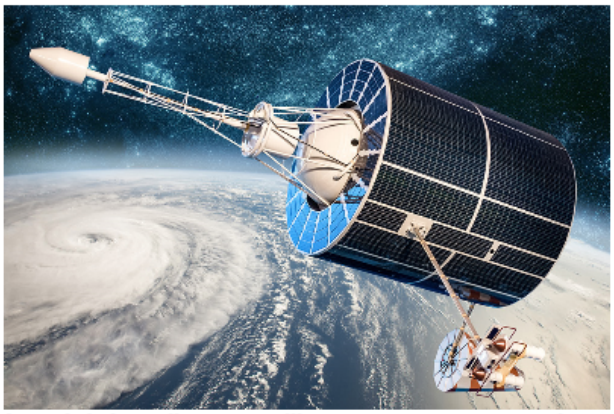
topology optimization process, which allows designers to remove materials without sacrificing part strength and integrity. The complex geometric structures that AM enables also yield more durable results, such as the improved strength-to-weight ratio found in lattice structures.



Lightweighting allows satellites to carry more equipment.

Rovers, satellites, and other objects that travel or carry items to space have strict weight limits. Lightweighting can allow them to carry far more equipment than a heavier version could because lightweighting spares more space for additional functional instruments and batteries. These add-ons can increase both functionality and powered-up time in space.

Adopting lightweighting processes requires additional employee training.

Despite its numerous benefits, lightweighting creates challenges for AM. Even if lightweighting may save on material costs, it requires up-front investment of time and money. Adapting production lines for lightweighting purposes can include purchasing new equipment and taking time to modify processes. As manufacturers introduce new lightweighting processes and materials, they must also provide new training for their employees and anticipate learning curve delays. Additionally, additive manufacturing processes may not be well suited to mass-scale production or building large components, both of which are important in industries like automotive and aerospace. Further, because AM technology is still evolving, part testing and inspection are not fully standardized, which can also slow mass production and make it difficult to build large parts safely.

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Lesson: 10/10

Review: Lightweighting with AM

- **DFAM**: The method of planning, testing, and building an additively manufactured part of optimal function.
- **Lattice structure**: Made of repeated unit cells.
- **Inspection**: The examination of a part during or after its creation to confirm that it adheres to specifications.
- **Topology optimization**: A mathematical method that optimizes material layout within a given design space.
- **Material reduction**: Decreasing the amount of a substance required for a build.

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Class Vocabulary

Additive Manufacturing	AM. The process of successively layering materials to make an object based on a three-dimensional computer model. Additive manufacturing allows for rapid prototyping, mass customization, and increased part complexity.
AM	Additive manufacturing. The process of successively layering materials to make an object based on a three-dimensional computer model. AM allows for rapid prototyping, mass customization, and increased part complexity.
CAD Software	Computer programs used to create a digital model of a part before it goes into prototyping or production. CAD software creates digital models that are converted to a format used by additive manufacturing machines.
Carbon Fiber	A material consisting of thin, strong crystalline filaments of carbon. Carbon fiber is a popular substitute for heavier metals in lightweighting.
Carbon Footprint	The amount of carbon dioxide and other carbon compounds emitted due to the consumption of fossil fuels by a particular person, group, or other organization. Entities often seek to reduce their carbon footprint for environmental reasons or to meet regulations.
Continuous Improvement	An ongoing effort to make products, services, or processes better. Continuous improvement identifies opportunities for improvement, takes action to achieve this, analyzes the results, and implements workable solutions.
Density	Mass per unit volume. Material density is a consideration of lightweighting, which aims to create parts of lower density but equal strength.
Deposition Head	A nozzle that melts and dispenses material. Deposition heads are used in directed energy deposition processes.
Design For Additive Manufacturing	DFAM. The methodology of planning, testing, and creating an additively manufactured part that functions optimally. Design for additive manufacturing allows engineers to focus on part functionality.
Directed Energy Deposition	DED. An additive manufacturing method in which focused thermal energy is used to melt materials as they are deposited on a build platform. Directed energy deposition is often used with metal powder or wire.
Fuel Efficiency	The capacity of an engine, especially that of a vehicle, to obtain energy from fuel. Fuel efficiency is often a desired outcome of lightweighting.
Image Projection Module	An assembly of components that focuses and directs the movement of a laser or electron beam in a specific pattern. Image projection modules are used during powder bed fusion processes.
Inertia	The tendency that objects have to resist changes to their state of motion. Inertia causes an object to either remain at rest or in its original state of motion until it is acted upon by an outside force.
Inspection	The examination of a part during or after its creation to confirm that it adheres to specifications. During inspection, defects may be identified and corrected.
Laser	An intense beam of light that can be precisely aimed and controlled. Lasers are sometimes used to selectively solidify or combine materials in additive manufacturing methods, including directed energy deposition and powder bed fusion.
Lattice Structures	A repeating, symmetrical pattern of crossing strips of material that leave diamond- or square-shaped gaps between them. Lattice structures provide excellent strength to a part.
Learning Curve	The rate of a person's progress in gaining experience or new skills. A steep learning curve can increase labor costs when new technologies or processes are adopted.
Lift Force	The sum of all the forces on a body that force it to move perpendicular to the direction of flow. Lift force allows aircraft to fly.
Lightweighting	In manufacturing, reducing the weight of a part or product without altering its function.

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Lightweighting	In manufacturing, reducing the weight of a part or product without altering its function. Lightweighting results in parts and products that are less heavy than previous iterations.
Loads	The opposition to applied force, such as a weight to be carried or moved. Designers must consider the loads a part will bear.
Machining	A subtractive manufacturing process that involves removing material to form an object. Machining includes methods such as milling, turning, and drilling that remove metal using cutting tools.
Mass	The amount of matter contained within an object. Mass gives an object weight when it is acted upon by gravity.
Material Jetting	An additive manufacturing method in which droplets of build material are selectively deposited onto a build platform. Material jetting uses ultraviolet light to cure photopolymer build material.
Material Reduction	Decreasing the amount of a substance required for a build. Material reduction results in a lighter part or product.
Material Substitution	Swapping one substance for another in the manufacture of a product. Material substitution of a lighter substance for a heavier one is a popular method of lightweighting.
Melt Pool	The small area of molten material that forms during directed energy deposition. Melt pools form a permanent material layer when they cool.
Part Design	The consideration of how a single part should best be designed geometrically. Part design is a kind of design for additive manufacturing.
Photopolymer	A thermoset polymer that cures and hardens when exposed to light. Photopolymers are used in material jetting.
Powder Bed Fusion	PBF. An additive manufacturing method that uses thermal energy to either sinter or melt together layers of powdered polymer, metal, ceramic, or other material. Powder bed fusion processes often use either lasers or electron beams as thermal energy sources.
Print Heads	The moving component on a printer that holds and distributes the part build material. Additive manufacturing machines, such as material jetting or binder jetting, have print heads.
Prosthetics	An artificial body part. Prosthetics can be made more flexible and durable by lightweighting methods.
Simulation Software	A program that allows the user to observe an operation without actually performing that operation. Simulation software is used widely to design equipment so that the final product will be as close to design specifications as possible without expensive modification.
Strength-To-Weight Ratio	The relationship between a material's strength and its weight. Materials that are light but also very strong have a high strength-to-weight ratio.
Stress	A force that attempts to deform an object. Common forms of stress include compression, shear, and tensile.
Substrate	A surface or medium that serves as a base for other materials or components. Some directed energy deposition processes deposit materials onto substrates rather than a build platform.
System Design	The consideration of what should be manufactured using additive manufacturing and what the component boundaries should look like. System design is a kind of design for additive manufacturing.
Thermal Energy	Energy in the form of heat. Thermal energy is used to fuse materials together in some additive manufacturing methods, such as directed energy deposition and powder bed fusion.
Thrust	A force that occurs when an object accelerates in one linear direction. Thrust can be measured in pound-force or newtons.
TO	Topology optimization. A mathematical method that optimizes material layout within a given design space for a given set of loads, boundary conditions, and constraints with the goal of maximizing the performance of the system. TO is a potent tool for lightweighting.
Tolerances	An unwanted but acceptable variation or deviation from a desired dimension of a part. Within a given range of tolerances, a part will still meet specifications.
Topology Optimization	TO. A mathematical method that optimizes material layout within a given design space for a given set of loads, boundary conditions, and constraints with the goal of maximizing the performance of the system. Topology optimization is a potent tool for lightweighting.

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Ultraviolet Light UV light. A potentially harmful wavelength of light that is below, or shorter, than violet on the light spectrum. Ultraviolet light is used to selectively solidify liquid photopolymer in material jetting.

Uniform Lattice A structure with identical unit cells repeated in every direction.

Unit Cells The smallest group of space in a material that constitutes a repeating pattern. Repeated unit cells make up a lattice structure.

Variable Lattice A structure with unit cells of varying size or spacing in different directions.

Wind Turbines A machine for generating electricity using a large, vaned wheel rotated by the wind. Wind turbines can be made more efficient by using lightweighting.

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